Implementation and Optimization of Receiver-Driven LEDBAT (rLEDBAT) within the NS-3 Simulation Framework

CSE 322: Computer Networks Sessional

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The objective of this project is to implement and optimize the Receiver-Driven Low Extra Delay Background Transport (rLEDBAT) protocol in the Network Simulator 3 (ns-3) building upon the theoretical framework of **RFC 9840** [1].

1 Analysis of the Base Protocol: rLEDBAT

The rLEDBAT protocol is a receiver-side adaptation of the Low Extra Delay Background Transport (LEDBAT) algorithm (RFC 6817). It is categorized as a "Less-Than-Best-Effort" (LBE) or "scavenger" transport mechanism, designed to utilize available slack bandwidth for background tasks (e.g., software updates) without interfering with high-priority, latency-sensitive traffic such as VoIP or gaming.

1.1 Core Operational Logic

Unlike traditional sender-side LEDBAT, which requires modifications to the server's TCP stack, rLEDBAT is implemented entirely at the receiver end. It controls the sender's transmission rate by modulating the standard TCP Receive Window ($rwnd$) field in the acknowledgment (ACK) headers.

The protocol follows a four-step cycle:

1. **Delay Measurement:** The receiver measures the Round-Trip Time (RTT) or One-Way Delay (OWD) using the TCP Timestamp option.
2. **Base Delay Tracking:** It maintains a running minimum of the delay ($base_delay$) observed over a moving window to estimate the zero-congestion propagation delay.
3. **Queuing Delay Estimation:** The current queuing delay is calculated as:

$$qdelay = current_delay - base_delay \tag{1}$$

4. **Window Modulation:** A specialized window value, W_{ledbat}, is updated using a proportional control loop. If $qdelay$ is below a target threshold $TARGET$ (default 100ms), the window increases; if $qdelay > TARGET$, the window decreases to reduce congestion.

The final advertised window sent to the sender is:

$$rwnd_{advertised} = \min(W_{ledbat}, rwnd_{flow_control}) \tag{2}$$

where $rwnd_{flow_control}$ is the standard flow control window.

2 NS-3 Implementation Strategy

The base implementation for this project will utilize the existing `TcpLedbat` class and acknowledgment generation logic.

2.1 Validation Baseline

The initial phase will involve validating the base rLEDBAT implementation against standard TCP CUBIC and Reno flows in a dumbbell topology. Key metrics to be evaluated include:

- **Yielding Ability:** Ensuring the rLEDBAT flow backs off as soon as a CUBIC flow enters the bottleneck.
- **Throughput Efficiency:** Measuring link utilization when no competing traffic is present.

3 Proposed Modifications and Technical Justification

To satisfy the requirement for protocol improvement, the following three modifications are proposed to address rLEDBAT's limitations in wireless and AQM-enabled networks.

3.1 Modification 1: Channel-Aware Delay Filtering (CADF)

Problem: In wireless networks (Wi-Fi/5G), link-layer retransmissions due to interference cause transient RTT spikes that are unrelated to network congestion. Standard rLEDBAT misinterprets these as congestion, causing unnecessary throughput reduction.
Proposed Improvement: Implement a median filter or a delay-trend filter within the `tcp-ledbat.cc` measurement logic. The receiver will only trigger a window decrease if a sustained increase in queuing delay is detected over multiple RTT samples.
Justification: Filtering the delay signal reduces the "false back-off" effect in noisy environments, improving background transfer speed without losing the LBE property.

3.2 Modification 2: Explicit Congestion Signaling (ECS) Integration

Problem: Modern AQMs (like CoDel or PIE) maintain extremely short queues. Because rLEDBAT reacts to a 100ms delay target, it may never "see" congestion if the AQM keeps the queue at 15ms, causing it to become too aggressive until the AQM is forced to drop packets.
Proposed Improvement: Modify the receiver's ACK handler to detect the ECN-Echo (ECE) flag. If an ECN signal is received, the protocol will immediately perform a multiplicative decrease of W_{ledbat}, bypassing the delay-based calculation.
Justification: ECN integration allows rLEDBAT to "hear" congestion that it cannot "see" via delay, ensuring fair coexistence with modern routers.

3.3 Modification 3: Adaptive Target Delay (ATD) Scaling

Problem: rLEDBAT uses a static target delay ($TARGET = 100ms$). On high-speed, low-latency links (e.g., 5ms base RTT), adding 100ms of delay is excessive and inefficient.
Proposed Improvement: Replace the static $TARGET$ with a dynamic target:

$$TARGET_{adaptive} = \max(base_delay \times \alpha, TARGET_{min}) \tag{3}$$

where α is a scaling factor (e.g., 0.2) based on the path's propagation characteristics.
Justification: Scaling the target delay ensures that the protocol's latency footprint is proportional to the path capacity, improving responsiveness on modern high-speed infrastructure.

4 Performance Metrics

To comprehensively evaluate the proposed modifications to rLEDBAT, the following performance metrics will be used to assess throughput efficiency, delay characteristics, reliability, and fairness across heterogeneous network conditions.

4.1 Throughput Evaluation

Throughput measures the efficiency of bandwidth utilization by the rLEDBAT flows without starvation of primary traffic.

- **Network Packet Rate (Packets/Second):** This is the total number of packets successfully delivered across the entire network divided by the simulation time. It captures the overall network efficiency.
  - *Modification Impact:* CADF should increase this rate in wireless scenarios by preventing the protocol from entering a "false back-off" state when RTT spikes are caused by link-layer noise rather than congestion.
- **Individual Node Throughput:** Measured as the received bits/bytes per second for a specific flow. This is critical for confirming that rLEDBAT maintains its "Less-Than-Best-Effort" (LBE) status.
  - *Modification Impact:* ATD Scaling will improve individual node throughput on low-latency paths by setting a more aggressive (but still safe) target delay, allowing the flow to better utilize the available capacity.

4.2 End-to-End (E2E) Delay

E2E delay is the average time taken for a packet to travel from the source node to the destination sink.

- **Evaluation Logic:** This metric verifies that the protocol successfully maintains its target delay (e.g., 100ms or your adaptive target).
- **Modification Impact:** ATD Scaling explicitly optimizes this metric by reducing the self-inflicted queuing delay on high-speed links where a static 100ms target would be unnecessarily high.

4.3 Jitter (Delay Variation)

Jitter is the absolute difference in end-to-end delay between consecutive packets of a single flow:

$$Jitter_i = |Delay_i - Delay_{i-1}| \tag{4}$$

- **Evaluation Logic:** Low-priority background traffic should ideally produce stable, low-jitter streams to avoid destabilizing the queues used by foreground applications.
- **Modification Impact:** CADF directly reduces jitter by filtering out transient wireless retransmission spikes, ensuring the congestion window ($RLWND$) does not oscillate erratically due to link-layer noise.

4.4 Network Delivery and Drop Ratios

These ratios measure the reliability of the protocol and the stress it places on the bottleneck buffer.

- **Packet Delivery Ratio (PDR):**

$$PDR = \frac{\text{Total Received Packets}}{\text{Total Sent Packets}} \tag{5}$$

- **Packet Drop Ratio:**

$$\text{Drop Ratio} = \frac{\text{Total Lost Packets}}{\text{Total Sent Packets}} \tag{6}$$

- **Modification Impact:** ECS Integration improves the drop ratio in networks with Active Queue Management (AQM). By reacting to ECN-Echo flags, the receiver can reduce its rate before the AQM (like CoDel or PIE) is forced to actually drop packets to signal congestion.

4.5 Fairness (Jain's Fairness Index)

To measure how equitably resources are shared among multiple competing flows, Jain's Fairness Index will be used:

$$J(x_1, x_2, \ldots, x_n) = \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n \sum_{i=1}^{n} x_i^2} \tag{7}$$

where n is the number of flows and x_i is the throughput of the ith flow.

- **Evaluation Logic:** This index ranges from $1/n$ (worst case) to 1 (perfect fairness).
- **Modification Impact:** All modifications aim to improve this index by reducing the "latecomer advantage," where new flows measure high base delays and starve existing flows. Specifically, **ATD Scaling** ensures that all rLEDBAT flows operating on the same path converge toward the same proportional delay target.

5 Expected Outcomes

The modified rLEDBAT implementation is expected to demonstrate:

- **Increased Wireless Robustness:** Higher throughput in scenarios with simulated 802.11 interference compared to the base implementation.
- **Improved AQM Compatibility:** Lower packet loss rates when running through `PieQueueDisc` or `CodelQueueDisc` bottlenecks.
- **Enhanced Fairness:** Reduction in the "latecomer advantage" through better base-delay estimation and adaptive thresholding.

This project will provide a comprehensive evaluation of receiver-driven congestion control, contributing a refined, more deployable LBE protocol to the ns-3 community.

References

[1] RFC 9840 - rLEDBAT: Receiver-Driven Low Extra Delay Background Transport, https://datatracker.ietf.org/doc/html/rfc9840, accessed January 25, 2026.